

Fire And Smoke Detection Using EfficientNET V2-B

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Abstract

This research presents a deep learning-based image classification system for early fire and smoke detection aimed at improving safety and minimizing damage. Unlike traditional sensor-based systems that often respond late and produce false alarms, this approach leverages image analysis for faster and more reliable detection. A convolutional neural network built on EfficientNetV2 using transfer learning achieves high accuracy while maintaining low computational cost. A custom dataset consisting of fire, smoke, and normal images was carefully prepared, cleaned, and enhanced to perform effectively under varying lighting conditions and environments. The model processes resized images as input and outputs probability scores for each class. It is integrated into an interactive Streamlit application, allowing users to upload images, view predictions, and analyze confidence values. Experimental results demonstrate high accuracy and fast response time, and when compared to traditional sensor systems and complex object detection models, this classifier performs more efficiently with lower computational requirements. Its lightweight architecture enables deployment on low-power CPUs and embedded devices, making it suitable for real-world fire safety applications, with future improvements focusing on video monitoring, edge deployment, and integration with IoT-based safety systems.

1. Introduction

Fire hazards are a serious danger to people, property, and the environment, so early detection is very important. Traditional systems like heat and smoke sensors often respond late, cover limited areas, and give false alarms in complex situations.

Recent progress in deep learning and computer vision has made fire detection faster and more reliable using image analysis. Convolutional Neural Networks can easily find visual features like flames, smoke, and scene details. Lightweight models like EfficientNet give high accuracy with low cost, making them good for real-time and edge use.

This research presents a deep learning system to classify images into Fire, Smoke, and Normal categories. It is built using TensorFlow and used through an interactive Streamlit app. The system uses image preprocessing, data augmentation, and transfer learning to improve performance. The model works well in different lighting conditions, complex backgrounds, and varying image quality. The goal is to create a practical, scalable, and low-cost fire detection system that gives early warnings and supports traditional sensor systems.

2. Objectives

The main objective of this research is to develop an accurate, efficient, and lightweight deep learning system to classify images into Fire, Smoke, and Normal categories. The study aims to solve the problems of traditional sensor-based fire detection by using faster and more reliable image-based analysis.

A key goal is to design and train a convolutional neural network based on EfficientNetV2 using TensorFlow, with transfer learning and fine-tuning to achieve high accuracy and low cost. This includes preparing the dataset, preprocessing, and data augmentation to improve performance in different conditions.

Another objective is to use the trained model in an interactive Streamlit application, where users can upload images, get predictions, and see confidence scores. The system is also tested using measures like accuracy, loss, and prediction confidence to ensure it works well.

Overall, the research aims to create a scalable and low-cost fire detection system, with future plans for video monitoring, IoT integration, and use on edge devices.

3. Literature Review

Fire and smoke detection has evolved from traditional sensor-based systems to vision-based methods. Conventional detectors, like heat and smoke alarms, often have limited coverage, slow response times, and false alarms, especially in complex settings. Early computer vision methods relied on handcrafted features such as color segmentation and motion analysis to detect

fire. These rule-based techniques could identify flames in controlled conditions but were very sensitive to changes in lighting, background clutter, and similar-looking objects. The rise of Convolutional Neural Networks has greatly improved fire detection by automatically learning features like flame shape, smoke texture, and scene context. Deep models such as VGG, InceptionV3, and ResNet achieved better accuracy, particularly when used with transfer learning. Object detection models like YOLO allowed real-time fire localization in videos, but their high computational cost and need for annotations limited their application in lightweight systems. Recent hybrid models with attention mechanisms have further enhanced performance but remain resource-intensive. To overcome these challenges, efficient architectures like EfficientNet and EfficientNetV2 have attracted interest for offering high accuracy with fewer parameters. This makes them suitable for real-time and low-power fire detection applications.

4. Existing System

The current systems of fire detectors primarily rely on heat, smoke, flame, and ionization sensors to detect fire by physical changes in the environment. These systems are also very common but tend to be slow, limited in range, and are also sensitive to false alarms due to dust, steam, or humidity.

The initial vision-based approaches relied on image processing algorithms like color thresholding, motion detection, and optical flow to identify fire and smoke. These rule systems proved useful in controlled environments, but were very sensitive to changes in lighting, shadows, and similar appearing objects.

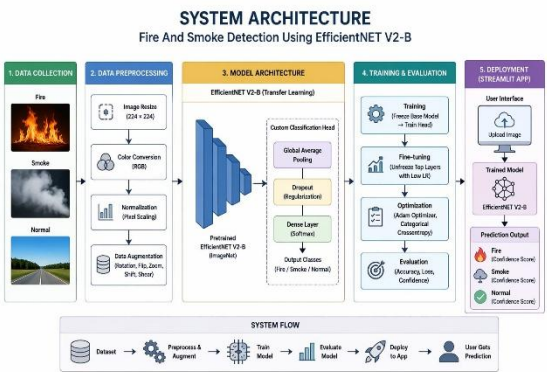
Fire detection was enhanced through deep learning as it allowed automatic feature extraction. MobileNetV2 is a lightweight model that lowered the cost of computation and could be deployed to the edge, although

it could occasionally fail on challenging visual patterns.

More advanced models such as EfficientNet and EfficientNetV2 improved both accuracy and efficiency. Although YOLO-based object detection offers real-time localization, it requires more resources and complex annotations. These limitations highlight the need for a lightweight, accurate, and scalable EfficientNetV2-based fire and smoke classification system.

5. Methodology

The methodology proposed has a systematic pipeline that generates a lightweight and correct fire and smoke image classification system. First, a wide range of data (Fire, Smoke, and Normal images) was gathered using open sources and repositories. The dataset was sorted into class-based folders by hand, purged of duplicate or poor-quality pictures, and divided into training and validation sets, to assure good performance assessment.



In preprocessing, all images were resized to 224 x 224 pixels and converted to RGB to fit the input model requirements. Normalization of pixel values was done to enhance stability of training. Data augmentation methods like rotation, zoom, shifting, shear, and horizontal flipping have been used to increase the robustness and minimize overfitting.

The main model has been constructed based on transfer learning with an EfficientNetV2 backbone that has been started with pretrained ImageNet weights. A custom classification head was added, which consists of Global Average Pooling, Dropout to learn regularization, and a Dense layer with SoftMax activation to classify the images into Fire, Smoke, and Normal. The architecture provides a trade-off between accuracy and computational efficiency.

The training was conducted in two stages: Feature extraction using frozen backbone layers, and fine-tuning of upper layers with a smaller learning rate. EarlyStopping and ModelCheckpoint techniques were employed to enhance convergence and avoid overfitting.

The accuracy, validation loss, and class-wise confidence scores were used to evaluate the model. The trained system was lastly deployed with an interactive Streamlit app, allowing users to post images and see real-time predictions with confidence visualization, which makes the solution applicable to real-world fire safety applications.

6. Implementation

The suggested fire, smoke and normal image classification system was deployed in a highly organized procedure that emphasized on efficiency and real time application. The system was written in Python and developed the model with TensorFlow and created the user interface with Streamlit. Data handling, image processing and visualization was supported using libraries like NumPy, PIL and Matplotlib. Training was done on a GPU enabled environment and inference was successfully tested on solely CPU based systems.

The data was sorted into three folders, Fire, Smoke and Normal. Images were loaded with ImageDataGenerator, preprocessing,

augmentation, and mini-batches were created with TensorFlow. The images were all downsampled to 224 x 224 pixels and normalized to have the same input.

The main model was EfficientNetV2 using weights of ImageNet. Its default classification layers were stripped and a new head and replaced with a three-class prediction Dense SoftMax output layer, Global Average Pooling and a Dropout regularization layer.

Transfer learning was done using a two step strategy. The EfficientNetV2 backbone was not trained at all in the first stage, but only the classification head was trained. The second stage involved upper layers unfreezing and fining-tuning with a lower learning rate. To enhance training stability, EarlyStopping and ModelCheckpoint were used.

7. Model Implementation

A fire, smoke and normal image classification model was proposed to offer proper and efficient real-time detection. EfficientNetV2 is the main feature extractor in the system due to its high accuracy, low computation cost, and quick convergence. Its compactness allows it to be used in surveillance systems, IoT, and other low-power devices. To enhance the learning of the features and minimize the time of training the model was started with a set of pretrained ImageNet weights.

The original top layers were eliminated in order to adjust the architecture to this task and a custom classification head was implemented. This involved a Global Average Pooling layer to minimize features dimension, a Dropout layer of 0.3 to regularize and a Dense output layer of three neurons representing Fire, Smoke and Normal classes. Class probabilities were produced by using SoftMax activation.

Training of the model was done in two phases with transfer learning. Initially, EfficientNetV2 backbone was frozen and only the custom head was trained. After that, some layers were chosen and unfrozen followed by fine-tuning of the selected layers using a lower learner rate to capture finer details of fire and smoke. Adam and categorical cross-entropy loss were used to optimize the model. The training stability was enhanced by EarlyStopping and ModelCheckpoint. The trained model was implemented into an interactive Streamlit application to predict in real-time and show its confidence.

8. System Architecture

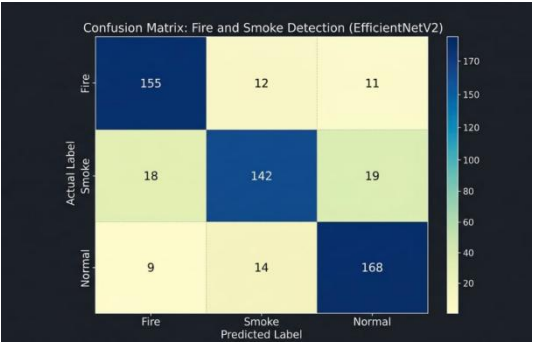
The system architecture proposed is aimed at providing effective processing of images, correct classification, and real-time applicability towards fire and smoke detection. It follows a modular pipeline consisting of image input, preprocessing, deep learning-based classification, and user interface deployment. The design is scalable, inexpensive in computation, and feasible to implement on surveillance and monitoring systems.

The architecture starts with the image input, whereby users enter fire, smoke or normal images via the training dataset or interactive Streamlit interface. The system is compatible with popular image formats like JPEG and PNG. At the preprocessing phase, images become 224 by 224 pixels, turned into RGB, normalized and made into arrays to fit the model.

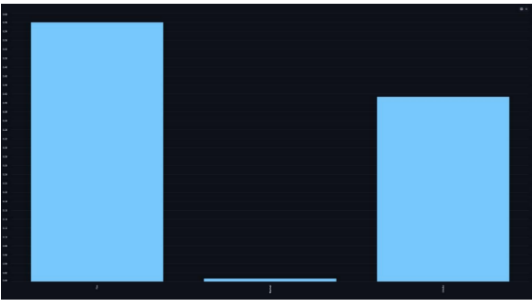
The classifier is EfficientNetV2 with a custom classification head. It also removes significant visual features and classifies as Fire, Smoke, or Normal. The results of the softMax activation can be interpreted as

class-wise probability scores. The model is small and capable of inference on CPU-only systems, and thus can be used on edge devices and embedded platforms.

With the help of the Streamlit interface, users can upload the images, make predictions and see class labels, confidence scores, and probability charts. Future extensions that are backed by the architecture include IoT alarms, integration of surveillance cameras and cloud-based monitoring to provide better applications of fire safety.



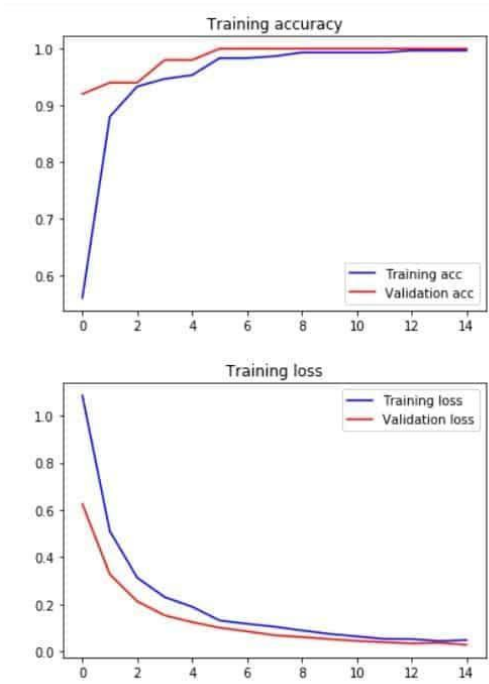
9.Result



Fig(2): Visualizing Prediction Confidence Using a Bar Graph

Index -- streamlit-generated	value
0 Fire	0.58
1 Normal	0.0068
2 Smoke	0.4132

Fig(3): Displaying Class-wise Prediction Probabilities



10. Application

The suggested fire, smoke, and normal image classification framework boasts of high-accuracy, quick inferences, and low-weight, which makes it have numerous applications in the real world. It can be incorporated in smart surveillance systems in order to have real time monitoring of fire and smoke via CCTV networks to enhance early warning and response.

In the industrial setting like factories, chemical plants and fuel storage facilities, the system can be used to identify smoke or flames when they are at an early stage before accidents occur. It is also applicable to embedded and IoT devices, which have limited computing needs.

The model has the potential to assist with the monitoring of forest fires with the use of drones, edge cameras, and remote stations to enable the authorities to detect wildfires in time. It may be used to supplement the conventional smoke sensors in intelligent buildings and smart homes in order to increase safety.

It is also applicable in transportation infrastructure like airports, tunnels and metro stations. The system may also help in research in fire safety, analyses of emergency responses, and eventual automated alerts, making it a versatile approach to various fire detection requirements.

11. Conclusion

The study introduces an effective and trustworthy system of fire, smoke, and normal image classification with EfficientNet V2 and transfer learning. The model is highly accurate and inexpensive to calculate and therefore is applicable in real time. The system is a part of an interactive Streamlit application that can be easily used to upload images and get a prediction and a visualization of the confidence. Generalization over a range of conditions

was enhanced by strong preprocessing, augmentation, and phased training. Its design is lightweight and can be used in surveillance, smart home, industries and IoT systems. In the future, video-based detection, temporal analysis, and automated alert integration will be incorporated into work to make it fire safe.

12. References

1. LeCun et al. explained the fundamentals of deep learning and CNNs for image recognition.
2. Krizhevsky et al. proved the effectiveness of deep CNNs for large-scale image classification.
3. Tan and Le introduced EfficientNet and EfficientNetV2 for efficient and accurate image classification.
4. Pan and Yang described transfer learning techniques used for pretrained model adaptation.
5. Kingma and Ba proposed the Adam optimizer for stable and faster model training.
6. Sharma et al. demonstrated CNN-based fire detection for improved safety monitoring.
7. Muhammad et al. presented early fire detection using deep learning methods.
8. Abadi et al. introduced TensorFlow for scalable deep learning implementation.
9. Streamlit enabled user-friendly real-time deployment of the system.
10. These studies collectively support the design, training, and deployment of the proposed fire and smoke detection model.